

# The impact of competition, trust and capital on renewable energy auction outcomes in sub-Saharan Africa: Analysing auctions in South Africa, Zambia and Namibia

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## ABSTRACT

Africa is short of power, despite having abundant renewable energy resources. Over the past decade, renewable energy auctions have emerged as an effective mechanism to competitively procure utility-scale private power projects. This paper identifies the elements contributing to efficient price and effective project realisation outcomes through comparative case studies in South Africa, Zambia and Namibia. The analysis combined existing literature and theory on those elements that contribute to success in Independent Power Projects (IPPs), with that on renewable energy auction design and implementation. The application of an integrated analytical framework shows that the introduction of renewable energy auctions in Africa provides an essential programmatic element which connects existing country and project level factors and is crucial in achieving superior project realisation and price outcomes, compared to projects procured through direct negotiation or feed-in tariffs.

Auction price outcomes are mainly determined by factors that act as auction signals, barriers and incentives primarily geared towards increasing competition in the procurement process and lowering projects' cost of capital. The case studies also show that project realisation is determined by bidder quality, project preparation levels, investor commitment, constraints (revenue stream and political economy) and post-award support. These are influenced by factors from all three levels – country, project and programme – that act as auction signals, barriers, incentives and support measures. Factors influencing bidders' trust in the auctioning authority and auction process are particularly important signals influencing both project prices and realisation rates.

## 1. Introduction

Sub-Saharan Africa needs more power, despite having considerable renewable energy resources (African Development Bank Group, 2017; Findt et al., 2014; IEA, 2014; The World Bank, 2017). Addressing this shortage requires considerable investment, from both public and private sources. Independent Power Projects (IPPs) – utility-scale (>5 MW) power projects that are built, owned and operated by the private sector – are the main source of private investment in power in sub-Saharan Africa. More than 280 of these projects have reached financial close since 1990, representing 20+ GW of installed capacity in over 30 countries (Alao and Kruger, 2020). The majority of these projects in the past 10 years have been renewable energy based.

Research into the factors determining both the prices (tariffs) and realisation rates (meaning successfully reaching commercial operation) of IPPs in sub-Saharan Africa has identified contributing elements to

success at country and project levels (Eberhard et al., 2016; Eberhard et al., 2017; Eberhard and Gratwick, 2011, 2013b; Gratwick and Eberhard, 2008b). Recent research emphasized the planning-procurement nexus as critical for increased IPP investment and sustainability (Eberhard et al., 2016), but does not analyse nor identify how a competitive procurement programme's design and implementation contributes to a project's ultimate success.

This paper aims to fill this empirical and theoretical knowledge gap by combining different bodies of literature to develop a comprehensive analytical framework that integrates programme-level factors, such as procurement design and implementation, with the country- and project-specific factors identified in the IPP literature (Table 1). This framework – a key contribution of the paper – is applied to the first sub-Saharan African countries that used renewable energy auctions and achieved remarkable outcomes – South Africa, Zambia and Namibia – to deepen our understanding of IPP outcomes in sub-Saharan Africa, and, more

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**Table 1**

Country and project level contributing elements of success for IPPs in Africa (adapted from Del Rio, 2017; Eberhard et al., 2016; Gephart & Ehrhart, 2018; Kreiss et al., 2016) (+ denotes positive impact on outcome; - denotes negative impact on outcome; ++ or – denotes comparatively stronger impact).

| Factors                                             | Detail                                                                                                                                                                                                                                   | Impact on Price                                                                                                                                         | Impact on Realisation                                                                                                                                                                                         |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Country level                                       |                                                                                                                                                                                                                                          |                                                                                                                                                         |                                                                                                                                                                                                               |
| Stability of economic and legal context             | Stability of macroeconomic policies (incl. repayment record & investment rating)<br>The legal system allows contracts to be enforced, laws to be upheld, and arbitration to be fair                                                      | (+) Stable investment context lowers investment risks, reducing cost of capital and increasing competition levels, leading to lower prices <sup>a</sup> | (+) Stability of investment context – especially judicial quality – improves contractual effectiveness and thereby realisation                                                                                |
| Energy policy framework                             | Energy policy framework allows private power investment<br>Market access through structured (preferably competitive) procurement programmes                                                                                              | (+) Clear energy policy that allows for private power investment reduces investment risks and increases competition levels, leading to lower prices     | (+) Clear energy policy that explicitly makes provision/allows for private investment ultimately is prerequisite for project realisation                                                                      |
| Regulatory transparency, consistency and fairness   | Transparent and predictable licensing and tariff framework (preferably included in bidding requirements/guidance)<br>Cost-reflective retail tariffs                                                                                      | (+) Good regulatory quality lowers investment risks, reducing cost of capital and increasing competition, leading to lower prices                       | (+) Good regulatory quality, especially on licensing and tariff approval, improve project realisation                                                                                                         |
| Coherent sectoral planning                          | Power-planning roles and functions clear and allocated to skilled, resourced, empowered planners<br>Fair allocation of new-build opportunities between utilities and IPPs<br>Plans are indicative rather than determinative              | (+) Coherent sectoral planning reduces investment risks, reducing cost of capital, and increases competition, leading to lower prices                   | (+) Coherent sectoral planning provides long-term investment signals, thereby improving project pipeline development, and as a result increasing the likelihood of project realisation                        |
| Competitive bidding practices                       | IPP procurement happens through competitive bidding programmes<br>Planning linked to timely initiation of adequately resourced competitive tenders/auctions – although auctions can be initiated outside planning framework if necessary | (+) Competitive bidding practices lower project prices                                                                                                  | (+) Efficient competitive bidding practices proactively provide the market with information required to shorten project development and negotiation timelines prior to bidding, thereby improving realisation |
| Programme level                                     |                                                                                                                                                                                                                                          |                                                                                                                                                         |                                                                                                                                                                                                               |
| Auction Design                                      |                                                                                                                                                                                                                                          |                                                                                                                                                         |                                                                                                                                                                                                               |
| Competition-inducing, scale-friendly auction volume | Auction volumes large enough to attract bidders (20 MW+), but small enough to ensure competitive pressure (especially                                                                                                                    | (++) Auction volume that is large enough to attract bidders, but smaller than offered volume, induces                                                   | (=) No theoretical expectation around impact on realisation                                                                                                                                                   |

**Table 1 (continued)**

| Factors                                                   | Detail                                                                                                                                                                                                                                                                            | Impact on Price                                                                                                                                                                    | Impact on Realisation                                                                                                                    |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Scale-friendly project size limits                        | in early rounds) – and preferably based on least-cost planning framework (Max) project size limits – if used – are based on clear market development, energy security criteria & enable economies of scale                                                                        | competition, leading to lower prices<br>(–) Project size caps limit economies of scale, increasing prices <sup>b</sup>                                                             | (+) Smaller projects are more quickly built, improving realisation                                                                       |
| Technology bands                                          | Initial auction rounds based on simple, tested technologies (e.g. solar PV)<br>Auction allows for adjustment of technology limits based on price and system optimisation considerations                                                                                           | (–) Technology bands limit competition, increasing prices                                                                                                                          | (=) No theoretical expectation around impact on realisation                                                                              |
| Bid award limits                                          | Bid award limits – if used – set to counter legitimate market domination or energy security concerns                                                                                                                                                                              | (–) Bid award caps limit economies of scale, increasing prices                                                                                                                     | (=) Bid award limits appear to have no discernible impact on realisation                                                                 |
| Bidding rounds                                            | Regular, scheduled bidding rounds increase competition and capture learning rates                                                                                                                                                                                                 | (++) Bidding rounds increase competition and induce market learning, lowering prices                                                                                               | (+) Bidding rounds increase project pipeline development and market learning, improving realisation <sup>c</sup>                         |
| Multi-criteria winner selection                           | Winner selection criteria transparent and rational<br>Price key (preferably only) determinant<br>Additional scoring criteria focus on socio-economic or system level impacts                                                                                                      | (–) Multiple bid scoring criteria dilute the importance of price and add additional cost drivers, increasing prices <sup>d</sup>                                                   | (=) Technical & SED scoring criteria appear to have no discernible impact on realisation                                                 |
| Price caps (unknown)                                      | Price caps (ceiling prices) undisclosed, if used                                                                                                                                                                                                                                  | (–) Undisclosed or no price caps discourage price anchoring, lowering prices                                                                                                       | (=) No theoretical impact expectation, although price caps being set too low can result in no project being awarded                      |
| Physical (project/bid preparation) qualification criteria | Physical (project/bid preparation) qualification criteria include, as a minimum, investor (lender) commitment<br>Late-stage qualification criteria coupled with long bidding timelines (for initial rounds) or site provision and preparation (incl. grid, permits) by auctioneer | (±) High physical qualification criteria increase bidding costs, lowering number of bids but increasing quality of bids (and if provided by auctioneer, can lower costs and risks) | (++) High physical qualification criteria (esp. lender commitment) improve project preparation and cost certainty, improving realisation |
| Financial (bid bond)                                      | Bid bonds set to ensure commitment of qualified bidders                                                                                                                                                                                                                           | (±) Bid bonds increase bidding costs, lowering                                                                                                                                     | (+) Bid bonds increase bidder commitment and                                                                                             |

(continued on next page)

Table 1 (continued)

| Factors                       | Detail                                                                                                                                                                                                    | Impact on Price                                                                                                 | Impact on Realisation                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| qualification criteria        | and the quality of competition                                                                                                                                                                            | the number of bids but increasing the quality of bids                                                           | cost certainty, improving realisation                                                                                                                     |
| Auction access requirements   | Auction access requirements – technical experience or financial status – set to limit participation to experienced, capable bidders                                                                       | (±) High auction access requirements lower number of bids, but improve quality of bids.                         | (+) High auction access requirements increase likelihood of capable bidders being awarded, increasing realisation                                         |
| Penalty regime                | Reasonable penalties (PPA reduction, performance bonds) incentivise project completion <sup>x</sup><br>Liquidated damages used to incentivise performance on non-technical aspects (e.g. SED commitments) | (–) Penalties increase project risks, increasing the cost of capital and other project costs, increasing prices | (+) Penalty regime incentivises project preparation and cost certainty, improving realisation                                                             |
| Staged bidding process        | Choice of one- vs. two stage bidding process balances transaction costs with time concerns                                                                                                                | (–) Staged bidding reduces competition, increasing prices                                                       | (=) No clear evidence of impact of staged bidding on realisation                                                                                          |
| Pay-as-bid, sealed bid format | Simple sealed bid, pay-as-bid auction format used to reduce bidder risks and increase competition levels                                                                                                  | (+) Simple bidding format and pricing rule increases competition, leading to lower prices                       | (+) Simple bidding format reduces winner's curse risk, improving realisation                                                                              |
| Auction Implementation        |                                                                                                                                                                                                           |                                                                                                                 |                                                                                                                                                           |
| Procurer                      | Procuring entity perceived as capable, benevolent and acting with integrity to build and maintain bidder trust                                                                                            | (++) Trust in the procuring authority increases competition, lowering prices                                    | (=) No evidence around impact on realisation                                                                                                              |
| Procurement process           | Bidding process is perceived as fair, transparent, secure to build and maintain bidder trust                                                                                                              | (++) Trust in bidding process increases competition, lowering prices                                            | (=) No evidence around impact on realisation                                                                                                              |
| Support to awarded projects   | Auctioneer assists awarded projects as honest broker to reach financial close and commercial operation                                                                                                    | (+) Auctioneer assistance improves likelihood of realisation, improving competition and lowering prices         | (+) Auctioneer support to awarded projects to financial close and COD improves realisation                                                                |
| Project level                 |                                                                                                                                                                                                           |                                                                                                                 |                                                                                                                                                           |
| Favourable equity partners    | Experienced (possibly international) firms with strong balance sheets compete to secure project contracts and realise projects<br>Local strategic shareholding encouraged                                 | (+) Strong, experienced equity partners can lower project costs and cost of capital, lowering prices            | (+) Experienced, strong equity partners able to navigate realisation risks, improving realisation<br>(+) Local strategic shareholders improve realisation |
| Favourable debt arrangements  | Competitive (low cost, long-tenored) debt – preferably                                                                                                                                                    | (+) Favourable debt arrangements                                                                                | (+) Debt from DFIs come with a halo effect,                                                                                                               |

Table 1 (continued)

| Factors                                        | Detail                                                                                                                                                                                                                                                                                                                                         | Impact on Price                                                                                                                      | Impact on Realisation                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|                                                | with considerable local capital to mitigate foreign exchange risk<br>Lenders review project contracts/ bidding documents to ensure bankability                                                                                                                                                                                                 | lowers the cost of capital, lowering prices                                                                                          | improving realisation<br>(+) Auction design in line with lender requirements improve realisation |
| Secure revenue stream                          | Secure revenue stream ensured through robust, bankable contracts (incl. international arbitration if necessary) with investment-grade utility (with full cost-recovery and high billing collection rates) or the use of risk mitigation and credit enhancement measures (e.g. sovereign guarantee, liquidity support, multilateral risk cover) | (++) Secure revenue stream reduces project risks, and thereby the cost of capital, and increases competition levels, lowering prices | (++) Secure revenue stream enables project to secure funding, improving realisation              |
| Strategic management and relationship building | Project bids include socio-economic development commitments – specifically strategic local shareholding (e.g. offtaker, empowerment shares)                                                                                                                                                                                                    | (–) Strategic socio-economic development commitments increase project costs & risks, increasing prices                               | (+) Strategic socio-economic development commitments improve project realisation                 |

<sup>a</sup> The impact of the investment context on pricing can be moderated through a well-designed, structured auction programme that has a secure revenue stream and is implemented by a trusted auctioning entity.

<sup>b</sup> The size of this impact seems to be somewhat limited by the fact that solar PV is a modular technology that appears to scale down relatively easily and cost-effectively (at least down to 20 MW).

<sup>c</sup> The overall impact of bidding rounds on realisation is nonetheless impacted by other external (economic, political) factors as well. It is often the path-breaking rounds or projects that appear to have an easier time getting built.

<sup>d</sup> The impact of this depends on the type and quantum of criteria. In general, SED criteria seem to have a more pronounced impact than technical criteria.

generally, in other challenging investment climates. As such, this paper is primarily a summarised comparative synthesis of these three in-depth case studies, each of which is (being) published stand alone journal articles ((Kruger, 2023, 2022; Kruger and Eberhard, 2019).

The rest of the paper is structured as follows: section 2 sets out the research design and methods, as well as the analytical framework used; section 3 introduces and summarises the case study countries; section 4 presents the results – including auction outcomes – and discusses the contributing factors, and section 5 presents the conclusions and policy implications.

## 2. Research design, methodology and analytical framework

### 2.1. Analytical framework

The analytical framework used during this research combines three bodies of literature:

- Research into the contributing elements of success of independent power projects (IPPs) at country and project levels;
- Research on (renewable energy) auction design; and
- Research on (renewable energy) auction implementation.

This was done to address a key gap in the IPP literature, which notes the importance of procurement but does not analyse how a procurement programme's design and implementation choices affect project price and investment outcomes. Likewise, the literature on renewable energy auction design tends to largely ignore the programme's broader context, the influence of project level factors and the impact of auction implementation choices. Together, these three bodies of literature provide an analytical framework that covers country, programme and project level contributing elements (Fig. 1).

Independent power projects<sup>1</sup> (IPPs) were introduced as part of the wave of power sector reforms that swept across the developing world,<sup>2</sup> beginning in the early 1990s, and which aimed mainly at improving sector performance and increasing power generation investment. Gratwick's (2007) Africa-focused analysis of IPPs explored the complementary aims and tensions of the main parties involved in the IPP transaction: investors were mainly looking for a reasonable return on their investment, more investment opportunities and the assurance that project survival would not be threatened by contracts unravelling (investment outcomes). Government was mainly interested in securing reliable electricity supply at a competitive (low) price (development outcomes). Both parties wanted to see projects reach financial close, get built and operate sustainably. The tension between these parties was mainly around agreeing on the tariff at which electricity would be sold by the IPP.

Gratwick's research found that the balancing of development and investment outcomes would ultimately lead to more sustainable projects. She also used country- and project-level analyses to empirically uncover the range of elements that contributed to success. Gratwick and Eberhard's (2008) classification of contributing elements to success at the country and project level has since been tested, refined and strengthened across a number of cases (Eberhard and Gratwick, 2013a, 2013b; Meyer et al., 2018; Eberhard et al., 2016; Eberhard et al., 2017).

At the project level, a secure revenue stream is essential for project success – that is, timely project realisation and competitive prices. Most IPPs in SSA are project financed, meaning that they are highly leveraged (using 70–90% debt) and that debtors only have recourse to the revenue stream – and not to the assets of the project sponsors (equity investors) – for servicing the debt (Eberhard et al., 2016). A secure revenue stream, captured in a contract – called a power purchase agreement (PPA) – between the offtaker of power (usually the state-owned utility company) and the power project is thus critical. The PPA stipulates the price per unit of energy (kWh) (or sometimes also for capacity, MW) to be paid over a fixed period (usually 15–25 years), and also allocates risks for a range of events and issues, including payment default, transmission delays, *force majeure*, and so on. For a PPA to be considered “bankable” – that is, meeting the requirements of debt providers – it needs to allocate risks to those best able to control and/or influence them (Gratwick and Eberhard, 2008).

Most sub-Saharan offtakers are not creditworthy, mainly owing to the utility's poor financial and operational performance as well as to consumer electricity tariffs not being cost-reflective (Trimble et al., 2016). This means that additional risk mitigation is needed for PPAs in sub-Saharan Africa to be bankable. This risk mitigation most often takes

the form of a guarantee from the state (sovereign guarantee) that it will fulfil the utility's payment obligations if it defaults. IPPs can also be provided with liquidity support – for example, the offtaker deposits a payment buffer, normally for 6 months' worth of payments, in a dedicated escrow account – and special political risk insurance to improve the risk profile of projects. A reasonable risk profile means that the cost of debt and equity (cost of capital) is also more likely to be reasonable and allows projects to charge more affordable tariffs (Eberhard et al., 2016).

It is, however, not merely the cost of capital, but also the providers thereof, that determines project success. The IPP literature shows that experienced, committed and development-oriented investors (both equity and debt providers) take a longer view and are more likely to help projects weather external and internal pressures, including through multi- and bi-lateral pressures on governments not to renege on contracts, with potential consequences for other country support programmes. This is one of the reasons why development finance institutions (DFIs) – such as the International Finance Corporation (IFC), FMO, KfW, Proparco and others – are such prominent investors in and lenders to IPPs on the continent.

At the country level, IPP success is influenced or determined by a handful of variables which include a stable investment environment encompassing factors such as political stability, economic growth prospects, sound macroeconomic policies, low levels of corruption and ease of doing business; clear (energy) policy that allows for private power investment accompanied by regulatory certainty on market access and electricity tariffs; and – importantly (because it is a factor that is often overlooked) – a dynamic, least-cost power expansion planning framework, based on realistic electricity demand scenario projections and up-to-date power generation costs translated into competitive procurement processes in a timely manner (Eberhard et al., 2016).

Where these country and project level elements are in place, one should see projects reaching financial close (FC) and commercial operation dates (COD) in relatively short time frames at competitive prices.

Given the recent emphasis of the planning-procurement nexus in the IPP literature (Eberhard and Gratwick, 2011; Eberhard et al., 2016), it is worth exploring the topic of the design and implementation of procurement programmes in greater detail.

A reverse auction – also called competitive procurement, internationally competitive bidding (ICB) or tendering – is a structured procurement programme (in contrast with direct negotiation) where contracts are awarded to power projects with the lowest tariff.<sup>3</sup> An auction is distinct from a feed-in tariff in that it is the market, and not an administrative agency, that sets project prices through a competitive process. The level of competition is determined by the number of bidders and also by the kinds of bidders: more technically and financially stronger bidders mean more competition.

More competition should translate into lower project prices through three mechanisms. The first is that bidders reveal more information about their real costs as they compete with each other in public bids, reducing information asymmetry between the private sector (developers/bidders) and the public sector (procurers) (Hubbard and Paarsch, 2016). Secondly, competition not only reveals costs but also encourages bidders to lower costs by squeezing margins in an attempt at improving their odds of winning through offering the lowest price (Eberhard and Naude, 2017; Haufe and Ehrhart, 2018). Thirdly, even if a procurer might have full transparency on a developer's costs in a direct negotiation (an unlikely but not impossible scenario), the procurer does not know if there might be someone better able to develop the project at a lower cost (Del Rio, 2017). A competitive procurement process provides the procurer with more certainty that they are getting the best

<sup>1</sup> IPPs are power generation plants built, financed and operated by the private sector, selling electricity to a publicly-owned or -backed utility offtaker through a long-term contract (Eberhard et al., 2016).

<sup>2</sup> The first IPP in Africa was built in Cote D'Ivoire in 1994, followed by Kenya in 1996 and Mauritius in 1997. They have since spread to more than 20 countries across the continent, representing more than 200 projects.

<sup>3</sup> Projects can also be scored on factors besides price, including project quality or socio-economic contributions, but in almost all cases the general weighting is in favour of price.

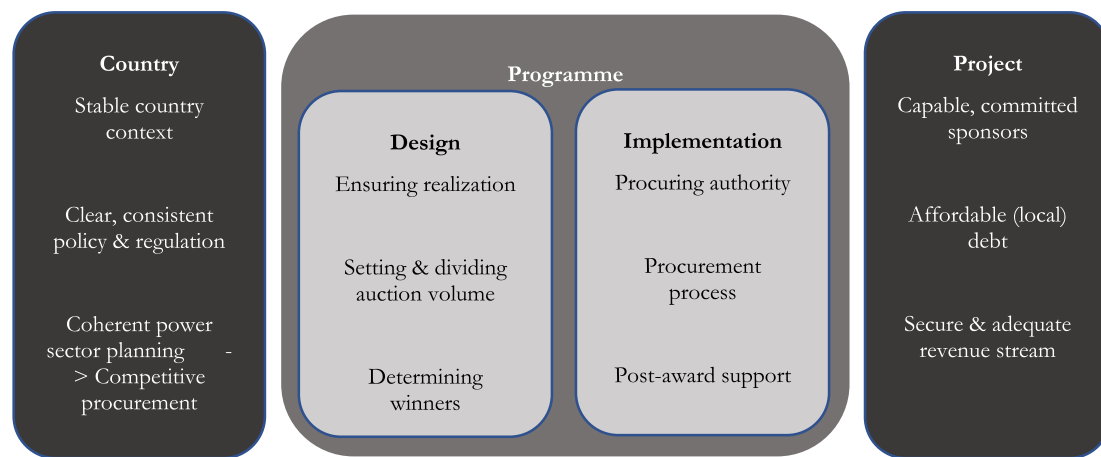


Fig. 1. Combining the country, programme and project level elements.

price that the market – and not only a single entity – has to offer. These benefits of competition can be undermined by low competition levels, as well as by collusion among bidders (a risk that increases with lower levels of competition) (Batz Liñeiro and Müsgens, 2023; Hubbard and Paarsch, 2016). A core objective of auction designers is therefore to maximise competition levels and reduce the likelihood of collusion.

Auction outcomes – mainly project prices and realisation rates – are influenced by both auction design and implementation factors.

Key issues in the design of auctions include:

- Auction volume: How much power should be procured? How is this determined? What is its impact on competition levels? How should the auction volume be divided between technologies, projects, bidders and over time?
- Winner selection: the criteria or rules used for bid-scoring, bid-ranking and awards; and
- Auction effectiveness: measuring the degree to which the procured projects get built on time, to a specified quality and with a specified impact.

The literature on renewable energy auctions has much to say on each of these three topics, drawing on economic theory (including game theory) and empirical evidence (Anatolitis et al., 2022; Del Río, 2017b, 2017a; del Río and Kiefer, 2023; Del Río and Linares, 2014; Diniz et al., 2023; Estache and Iimi, 2012; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; International Renewable Energy Agency, 2019; Kitzing et al., 2019; Kitzing and Weber, 2015; Kreiss et al., 2016; Kylili and Fokaides, 2015; Lucas et al., 2017, 2013; Mastropietro et al., 2014; Matthaus, 2020; Moreno et al., 2010; Welisch, 2018).

Key issues in auction implementation include:

- How bidders decide whether to submit a bid or not (bidding decision) (Chiu et al., 2010; Zitron, 2006). While the renewable energy auction literature discusses some of the factors underlying the bidding decision, we argue that the issue of trust is central to answering this question – trust in the bidding authority (auctioneer) and trust in the bidding process (Chiu et al., 2010; Lewicki et al., 2006; Liu et al., 2016; Mayer et al., 1995; Zaheer et al., 1998; Zitron, 2006); and
- How auctioning authorities ensure the effective implementation of procured projects through the coordination of public sector responsibilities and processes (implementation effectiveness) (Boulle et al., 2015; Eberhard et al., 2014; Montmasson-Clair and Ryan, 2014).

Together, these auction design and implementation elements constitute the programme level of analysis. In combination with the

country- and project-level factors identified in the IPP literature, this expanded framework (Table 1) serves to deepen our understanding of private power investment outcomes in Africa. The combined framework allows for deeper insights into, and a more comprehensive analysis of, the factors impacting auction outcomes.

## 2.2. Research design and methodology

We selected a qualitative, exploratory comparative case study-based research design that uses an abductive approach to theory development. Case studies are especially useful where phenomena, such as a government procurement programme, are not well understood and need to be studied within their context – and where the boundaries between the context and the phenomena are not clear (Babbie and Mouton, 2001; Kumar, 2011; Yin, 2003). Case studies – both single and multiple – are the dominant research design used throughout most of the literature on IPPs and renewable energy auctions. Multiple cases were selected based on the potential to “... draw otherwise inaccessible conclusions” (Creswell and Poth, 2018).

Cases selection was based on a “deviant case selection” logic (Seawnght and Gerring, 2008: 302): cases were selected because, with “... reference to some general cross-case relationship, (they) demonstrate a surprising value; they are poorly explained.” The three cases (South Africa, Zambia and Namibia) selected had all revealed surprisingly low prices for awarded projects, at or below the global Levelised Cost of Electricity for the technologies – prices that were not thought to be reflective of the general investment conditions in sub-Saharan Africa or these specific countries or their sectors. The purpose of deviant case analysis is “... to probe for new – but as yet unspecified – explanations” (Seawnght and Gerring, 2008: 302). The initial focus on the price outcomes of these auctions was supplemented by a focus on the project realisation outcomes, both as a “test” of the viability of these prices<sup>4</sup> and an important outcome in its own right, given the study’s foundational impetus to accelerate power generation investment in the region.

Dubois and Gadde (2002) explain the abductive approach in case study research to be a “... continuous movement between an empirical world and a model world. During this process, the research issues and the analytical framework are successfully reoriented when they are confronted with the empirical world”. It requires that the research starts with a preliminary analytical framework that is developed over time “... according to what is discovered through the empirical fieldwork, as well as through analysis and interpretation” (Dubois & Gadde, 2002: 555).

<sup>4</sup> The assumption being that if these prices were too low – unrealistic – these projects would not be able to be finance and built.



The paper seeks to develop plausible theoretical explanations by an iterative back-and-forth movement between data and theory, allowing for the possibility of exploring the applicability of *a priori* constructs and constant dialogue between the evidence and the enfolding literature (Eisenhardt, 1989; Ragin, 1999). This two-way process has, for example, led to the identification of additional bodies of theory and literature – not originally considered – that help explain the project outcomes.

Data was gathered in two primary ways: collecting and analysing documentation from the relevant case studies (including requests for qualification and proposals documents, contracts, project agreements, financing and credit enhancement offers, media reports, press statements, relevant policies and regulations, meeting minutes and scholarly publications) (see Appendix B) and 45 unstructured interviews (guided conversations) with officials from government agencies and departments (including procuring entities, utility representatives, energy ministries and regulators), the private sector (including successful and unsuccessful bidders and project developers, debt and equity providers, contractors) and advisors (legal, financial, technical, environmental, social) closely involved in the design and implementation of the renewable energy auction programmes in the case study countries (See Appendix A). The interviews were carried out in the 2018–2020 period.

A purposive sampling strategy (Saunders et al., 2016; Yin, 2018) was used to select interview participants, based on their involvement with and familiarity with the auction programmes. Most interviews were in-depth conversations taking anywhere between one and 3 h, in line with the fluid “intensive interview” method typically employed as part of the case study approach (Rubin and Rubin, 2012; Weiss, 1994). Interview notes were coded manually, mainly using theoretical concepts from the literature, but also allowing for the data to highlight important new themes. Case study drafts were shared with key informants for validation.

### 3. Introducing the cases

As the sub-Saharan African country with the most experience of the auction procurement framework, South Africa forms the first case study. In 2011, South Africa was the first country in sub-Saharan Africa to launch dedicated renewable energy auctions, called the Renewable Energy Independent Power Producers Procurement Programme (REI4P). The programme procured 112 projects, representing 6327 MW across a number of renewable energy technologies. The country has also witnessed a dramatic fall in project prices over four procurement rounds (2011–2015). South Africa’s auction programme is renowned for the way that it incorporates a range of detailed socio-economic and economic development criteria and requirements in the auction process. The last round of procurement was in 2015, but new procurement rounds were initiated in 2021.

Zambia was the second<sup>5</sup> country to run an auction in sub-Saharan Africa, through the International Finance Corporation’s (IFC) Scaling Solar programme in 2015/16. This was the first instance of this World Bank Group-packaged approach being used and included a range of stapled risk mitigation and credit enhancement measures. The programme also made concessionary finance packages available to bidders and emphasized speed, quality and price. The resulting bid prices were globally competitive at the time of announcement and quite controversial in the sub-Saharan region: several project developers felt that the prices were setting unrealistic market expectations due to what they perceived as a high level of concessional elements in the programme. Despite delays in reaching financial close, projects were still built in

good time and began commercial operation in early 2019. A second round of Scaling Solar procurement was cancelled in 2018/19 owing to the declining credit-worthiness of ZESCO – the Zambian electricity utility company – and the Zambian economy in general. Nevertheless, a parallel procurement programme – the Zambian version of the GETFIT solar facility – was launched in 2018 and resulted in even lower prices – although awarded projects have not been able to reach financial close.

Namibia was the third country to launch an auction programme, following closely on the heels of Scaling Solar in Zambia. It featured important differences from the other programmes in the region. While Namibia managed to secure a project at prices similar to those in Zambia’s Scaling Solar programme and South Africa’s REI4P, it did so without offering any risk mitigation or credit enhancement measures, and no concessionary finance. The auction was implemented by the national utility NamPower, which was also the offtaker of power and eventual part shareholder in the project. The programme furthermore featured no donor involvement or assistance. It also incorporated a unique and innovative debt arrangement between a commercial bank and a development finance institution. A second round of procurement was launched in 2019, and announced prices were again highly cost competitive.

## 4. Results and discussion

### 4.1. Reviewing auction outcomes

We compare auction outcomes across the following dimensions: price, project realisation and direct socio-economic development impacts.

Many have questioned the comparability of auction price outcomes because of the variance in country and project conditions, in contractual and financing arrangements and in bidder assumptions (Del Río, 2017b; Dobrotkova et al., 2018; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018). Consequently, we focus primarily on major trends of convergence and divergence between countries’ auction price outcomes for solar PV. Project prices are compared according to the year in which projects were expected to reach financial close, as it was assumed that bidders would base their technology cost assumptions on these dates.

Overall, the price outcomes show remarkable convergence across cases, as well as with the average global levelized cost of electricity (LCOE) for solar PV (Fig. 2) – despite initially expensive projects in South Africa’s earlier bidding rounds.

In general, the correlation between higher competition levels and lower-price outcomes (Batz Liñeiro and Müsgens, 2023; Gephart et al., 2017a; Hubbard and Paarsch, 2016) across the three cases appears to hold, with competition generally increasing over time and prices reducing (Table 2 and Fig. 3). A notable exception was South Africa’s fourth bid window (BW4), which saw lower levels of competition compared to previous windows. The reduction in the number of bids in BW4 masked a concurrent increase in the quality of bids – as measured by the experience, size, and financial strength of bidders – which effectively increased competition levels (despite a lower number of bids submitted), and consequently drove down prices. To be clear, competition is not the only factor determining price outcomes, and while an important aim of the paper is to further our understanding of how various factors might have influenced competition levels, we also aim to identify and understand other price determinants beyond competition.

All three cases featured high realisation rates (95%–100%) though there were some delays in reaching commercial operation. The GETFIT projects in Zambia, which have not been able to advance to financial close, have been an exception (discussed further below).

The following section will analyse the contributions of the country-, project- and programme-level elements to these outcomes – summarised for ease of reference in Table 2.

<sup>5</sup> While it could be argued that Uganda’s GETFIT solar facility was the second auction in the region, it is strictly considered a hybrid programme because it combines a feed-in tariff approach with a competitively set top-up premium payment. While interesting and potentially illuminating, the case was not included in the final analysis.

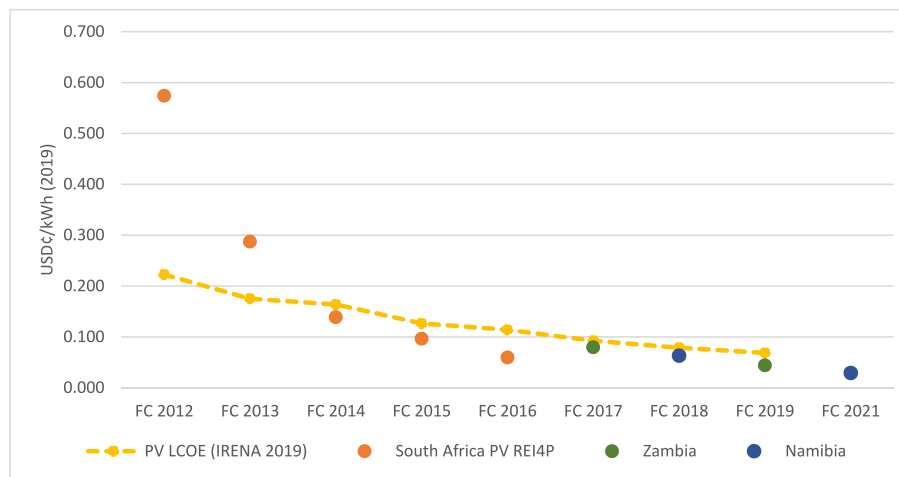


Fig. 2. Solar PV prices at expected financial close compared to global LCOE – South Africa, Zambia, Namibia.<sup>61</sup>

Table 2

Competition levels – South Africa, Zambia, and Namibia.

|              | EOI/<br>Qualification | Bids<br>submitted | Projects<br>awarded | Competition level (No.<br>of bids submitted/<br>awarded projects) |
|--------------|-----------------------|-------------------|---------------------|-------------------------------------------------------------------|
| South Africa |                       |                   |                     |                                                                   |
| BW1          | N/A                   | 53                | 28                  | 1.9                                                               |
| BW2          | N/A                   | 79                | 19                  | 4.2                                                               |
| BW3          | N/A                   | 93                | 17                  | 5.5                                                               |
| BW4          | N/A                   | 77                | 26                  | 3.0                                                               |
| Zambia       |                       |                   |                     |                                                                   |
| Scaling      | 48                    | 11                | 2                   | 5.5                                                               |
| Solar 1      |                       |                   |                     |                                                                   |
| GETFiT       | 41                    | 15                | 6                   | 2.5                                                               |
| Namibia      |                       |                   |                     |                                                                   |
| Hardap       | 250                   | 13                | 1                   | 13.0                                                              |
| Khan         | 100                   | 16                | 1                   | 16.0                                                              |

Table 3

Summary of auction effectiveness outcomes – South Africa, Zambia, and Namibia.

|              | MW<br>procured          | Technologies        | Realisation<br>Rate <sup>a</sup> | Average<br>Realisation<br>period (months<br>from award to<br>COD) |
|--------------|-------------------------|---------------------|----------------------------------|-------------------------------------------------------------------|
| South Africa |                         |                     |                                  |                                                                   |
|              | <b>6127<sup>b</sup></b> | Solar PV, Onshore   | 95%                              |                                                                   |
| BW1          | 1425                    | Wind, CSP, Biomass, |                                  | 38,6                                                              |
| BW2          | 1040                    | Landfill gas, Small |                                  | 32,8                                                              |
| BW3          | 1457                    | hydro               |                                  | 42                                                                |
| BW4          | 2205                    |                     |                                  | N/A                                                               |
| Zambia       |                         |                     |                                  |                                                                   |
|              | <b>219</b>              | Solar PV            |                                  |                                                                   |
| Scaling      | 89                      |                     | 100%                             | 19,5                                                              |
| Solar 1      |                         |                     |                                  |                                                                   |
| GETFiT       | 120                     |                     |                                  | N/A                                                               |
| Namibia      |                         |                     |                                  |                                                                   |
|              | <b>65</b>               | Solar PV            |                                  |                                                                   |
| Hardap       | 45 (37)                 |                     | 100%                             | 13                                                                |
| Khan         | 20                      |                     |                                  | N/A                                                               |

<sup>a</sup> Only considering projects for which scheduled COD deadlines had already passed. If the GETFiT and Khan projects fail to reach commercial operation, this will dramatically lower this figure.

<sup>b</sup> This does not include the projects procured through the Small IPP procurement programme (Kruger et al., 2021).

## 4.2. Country level elements

### 4.2.1. Favourable investment climate

The IPP literature posits that a favourable investment climate, which leads to both increased investor interest (thereby improving competition) and a lower cost of capital (through improved sovereign credit ratings), should ideally have a number of macroeconomic fundamentals in place – including low inflation rates, moderate national-debt levels, national wealth (GDP/capita), a large economy, and well-developed, deep local capital markets. Property rights, contract enforcement and the integrity of procurement processes should, furthermore, be protected by a strong, independent judiciary (Eberhard et al., 2016; Eberhard and Gratwick, 2011; Vaaler et al., 2008; Woodhouse, 2005).

The three countries offer, at best, a tenuous link between investment climate conditions and price outcomes (Kruger, 2022, 2023; Kruger and Eberhard, 2019). Auction price outcomes – impacted also by declining equipment costs globally – improved even as investment climate conditions worsened. All three countries, for example, saw average project prices decline over successive auction rounds at a time when macroeconomic conditions deteriorated – including slower, or even negative, economic growth levels (Fig. 4), credit rating downgrades, and steeply rising national debt levels (Table 4). In South Africa and Namibia, the prominence of local debt providers (thanks to Namibia's currency peg to the South African Rand) partly shielded the highly geared projects from exposure to higher borrowing costs on international markets as credit ratings declined (Fig. 5), thus strengthening the case for local capital market depth as contributing to project success (Eberhard and Gratwick, 2011).

Nevertheless, while auction efficiency (price outcomes) seems to have been somewhat shielded from the impacts of macroeconomic deterioration, auction effectiveness (including project realisation) has not. Economic shocks, rapid currency depreciation and long-standing economic mismanagement, for example, have derailed the financial close of the awarded GETFiT projects and have undermined medium-term prospects for private power investment in Zambia.

The above analysis confirms the importance, but also limitations, in relying solely on the investment climate as an explanatory factor for auctions' price and investment outcomes, necessitating further probing of other factors at the country, project and programme levels.

### 4.2.2. Energy policy framework and regulation

Beyond the investment climate, the context for accelerated and sustained IPP investment is further strengthened through an enabling policy and regulatory framework that should, at the very least, make space for the private sector's participation in power generation. It also ideally clarifies the roles of the private vs public sector entities, and

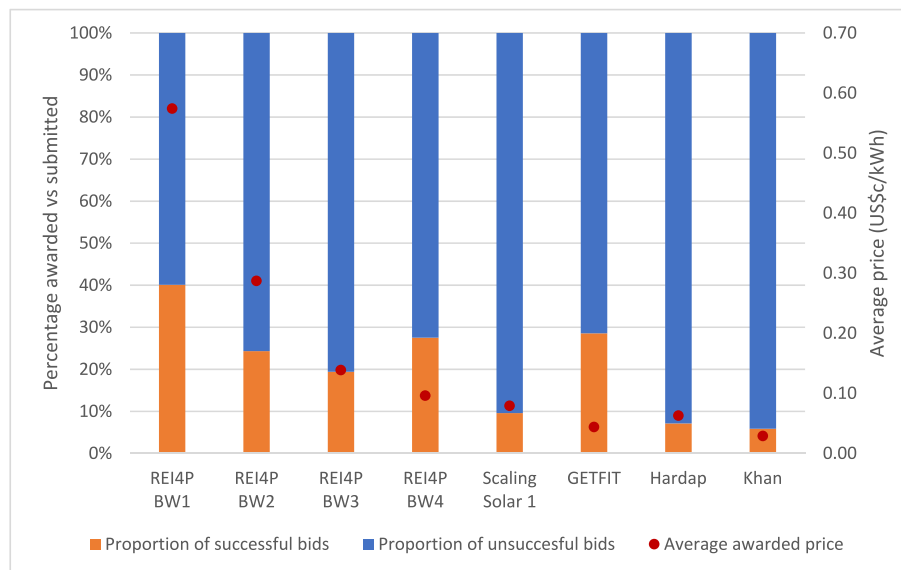


Fig. 3. Competition levels – MW submitted vs MW awarded: South Africa, Zambia, and Namibia.

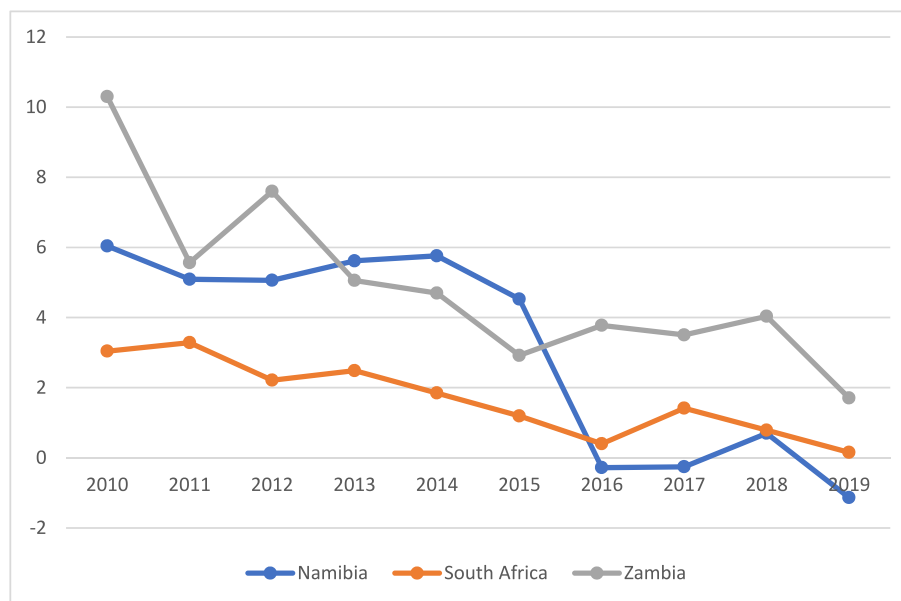


Fig. 4. Annual GDP growth rate, 2010–2019: South Africa, Zambia and Namibia (Source: [The World Bank Group, 2020](#)).

specifies how, and by whom, IPPs should be procured (Eberhard et al., 2017; Eberhard and Gratwick, 2011; Urpelainen and Yang, 2017). This enabling framework is further strengthened through policies that clarify governance arrangements for state-owned utilities and the presence of an independent sector regulatory agency that ensures the sector's financial health through setting cost-reflective consumer tariffs and implementing transparent, predictable generation-licensing and PPA approval mechanisms (Eberhard et al., 2016; Kapika and Eberhard, 2013).

All three cases illustrate that enabling policy and planning – while important – is not a sufficient condition for accelerating private

investment in power: they need to be complemented by structured procurement processes that provide a clear route to market (Kruger, 2022, 2023; Kruger and Eberhard, 2019).

#### 4.2.3. The planning procurement nexus

The contribution of the planning-procurement nexus to IPP outcomes is not as straightforward as originally assumed (Gratwick and Eberhard, 2008). Ideally, a regularly updated power sector expansion plan should project demand, establish a system reliability standard, select the technologies that can meet demand at the lowest cost (including alternative scenarios), and make clear the basis on which new-build opportunities are to be allocated to either the private sector or to state-owned utilities. The plan needs to be translated into international competitive bidding rounds on a timely and regular basis so as to be effective (Eberhard et al., 2017).

This ideal framework was not in evidence in any of the case study countries, and plans have generally failed to keep up with price and

<sup>6</sup> Bid prices have been adjusted for 2019 and exchange rates are based on the average exchange rate in the year that the bid was announced. Prices are shown for the year in which financial close was expected/required according to the bidding rules.



Table 4

Summarised country, programme and project level contributing factors in South Africa, Zambia and Namibia.

| Factors                                                               | South Africa                                                                                                     | Zambia                                                                                                                                                    | Namibia                                                                                                                                                                       |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Country level                                                         |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                                               |
| Stability of economic and legal context                               | Relative economic, political stability, although sovereign credit successively downgraded; judiciary independent | Rapidly deteriorating state finances, sovereign credit ratings downgrades into imminent default status; relatively low levels of judicial independence    | Relative economic, political stability, although sovereign credit successively downgraded; judiciary independent                                                              |
| Energy policy framework                                               | Energy policy allows for private power investment; market structure set out, but not implemented                 | Energy policy allows for private power investment; new policies see expanded role for renewables                                                          | Energy policy allows for private power investment; modified single buyer market enshrined in legislation                                                                      |
| Regulatory transparency, consistency and fairness                     | Regulatory quality deteriorating over time, despite regulatory independence                                      | Regulatory quality relatively highly rated, but regulatory independence undermined                                                                        | High levels of regulatory quality                                                                                                                                             |
| Coherent sectoral planning                                            | IRP provides coherent power generation expansion plan, but overly prescriptive and contested                     | No coherent power sector expansion plan                                                                                                                   | NIRP only officially adopted in 2019                                                                                                                                          |
| Competitive bidding practices                                         | Use of competitive bidding practices informed by IRP planning framework                                          | Use of competitive bidding practices for utility-scale solar IPPs                                                                                         | Switch to competitive bidding practices for utility-scale PV IPPs after FIT and DN project experiences                                                                        |
| <b>Programme level</b>                                                |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                                               |
| <i>Auction Design</i>                                                 |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                                               |
| Competition-inducing, scale-friendly auction volume                   | Large (1000 MW+) auction volumes from BW1 – slightly reduced for later rounds                                    | Modest auction volumes – ~100 MW                                                                                                                          | Modest to small auction volumes: 20–37 MW                                                                                                                                     |
| Scale-friendly project size limits                                    | Large project size limits (75 MW PV; 140 MW wind)                                                                | Modest project size limits (40–55 MW)                                                                                                                     | Small to modest project size limits: 20–37 MW                                                                                                                                 |
| Technology choice: neutral vs. specific                               | Technology specific demand bands in REI4P; technology neutral SPI4P                                              | Technology specific: solar PV                                                                                                                             | Technology specific: solar PV                                                                                                                                                 |
| Bid award limits                                                      | No bid award limits                                                                                              | Bid award limits                                                                                                                                          | Bid award limits (by default since only single projects auctioned)                                                                                                            |
| Bidding rounds                                                        | Regular, scheduled auction rounds announced from BW1                                                             | Auction rounds announced for Scaling Solar, but not realised; No schedule for GETFIT solar                                                                | No auction schedule                                                                                                                                                           |
| Winner selection                                                      | Price (70%) + Socio-economic development commitments (30%)                                                       | Price (100%): Scaling Solar<br>Price (80%) and Project preparation + SED criteria (20%): GETFIT                                                           | Price (70%) and Technical (30%): Hardap<br>Price (80%) and SED (20%): Khan                                                                                                    |
| Price caps                                                            | Announced price caps from BW1-3; removed for BW4                                                                 | No price caps                                                                                                                                             | No price caps                                                                                                                                                                 |
| Physical (project/bid preparation) qualification criteria             | Late-stage physical qualification criteria, including investor commitment                                        | Scaling Solar: site provided & prepared, but lender commitment required<br>GETFIT: early stage physical qualification, but with lender signed term sheets | Sites provided and prepared. Investor commitment letters required                                                                                                             |
| Financial (bid bond) qualification criteria                           | Relatively expensive bid bonds required                                                                          | Expensive bid bonds required for both Scaling Solar and GETFIT                                                                                            | Modest bid bond required for Hardap. No bid bond for Khan                                                                                                                     |
| Auction access requirements                                           | Only analysed financial track record or assets;<br>Heads of EPC and O&M contracts                                | Project development experience and financial standing; Heads of EPC & O&M contracts                                                                       | Only analysed project development experience (modest level); Heads of EPC and O&M contracts only for Hardap                                                                   |
| Penalty regime                                                        | Performance bonds; PPA reduction clause; Liquidated damages only for SED performance                             | Performance bond (Scaling Solar); PPA reduction clause; Liquidated damages for delays (and underperformance in Scaling Solar)                             | Performance bonds (Hardap only); PPA length reduction clauses; Liquidated damages for delays and underperformance (Hardap only)                                               |
| Staged bidding process                                                | Single stage                                                                                                     | Two-stage                                                                                                                                                 | Single stage                                                                                                                                                                  |
| Pricing rule, auction format                                          | Sealed bid, pay-as-bid                                                                                           | Sealed bid, pay-as-bid                                                                                                                                    | Sealed bid, pay-as-bid                                                                                                                                                        |
| <b>Auction Implementation</b>                                         |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                                               |
| Procurer                                                              | IPP office viewed as capable, acting with integrity                                                              | IFC (through IDC) and GETFIT secretariat viewed as capable, acting with integrity                                                                         | Generally positive view of NamPower as procurer, but tainted by award controversies                                                                                           |
| Procurement process                                                   | REI4P viewed as fair, transparent, secure                                                                        | Both Scaling Solar and GETFIT viewed as fair, transparent, secure                                                                                         | Hardap tainted by 3 × 10MW award controversy, reducing perceptions of award fairness, transparency; Khan process viewed in better light, but concerns about indexation option |
| Support to awarded projects                                           | IPP office supported projects to reach financial close, COD                                                      | IDC/IFC provided little support to awarded projects; GETFIT secretariat supporting, but unable to reach FC                                                | NamPower supported Hardap to reach FC, COD;                                                                                                                                   |
| <b>Project level</b>                                                  |                                                                                                                  |                                                                                                                                                           |                                                                                                                                                                               |
| Favourable equity partners                                            | Mainly large international firms as lead bidders (with local shareholders) – especially later rounds             | Large to moderate international firms awarded                                                                                                             | International firms with moderate experience awarded                                                                                                                          |
| Favourable debt arrangements                                          | High gearing with long-term commercial debt (especially later BWS)                                               | High gearing with long-term, concessionally priced DFI debt                                                                                               | High gearing from commercial debt, with DFI support to enable 15 year tenor                                                                                                   |
| Creditworthy off-taker                                                | Eskom not credit-worthy off-taker                                                                                | ZESCO not credit-worthy off-taker and in default on IPP and other payments                                                                                | NamPower credit-worthy off-taker                                                                                                                                              |
| Secure and adequate revenue stream                                    | PPA and other contractual provisions bankable, secure                                                            | PPA and other contractual provisions bankable, secure                                                                                                     | PPA and other contractual provisions bankable, secure                                                                                                                         |
| Credit enhancements and other risk management and mitigation measures | Sovereign guarantee                                                                                              | Sovereign guarantee, liquidity support and (for Scaling Solar) multilateral termination support                                                           | No official risk mitigation or credit enhancement (though DFI loan guarantee provided to Standard Bank)                                                                       |
| Strategic management and relationship building                        | Substantial socio-economic development commitments, incl. empowerment shareholding                               | IDC is minority shareholder in Scaling Solar projects; some SED commitments from GETFIT projects                                                          | NamPower is minority shareholder in Hardap, + PDN shareholding; PDN shareholding + SED commitments for Khan                                                                   |

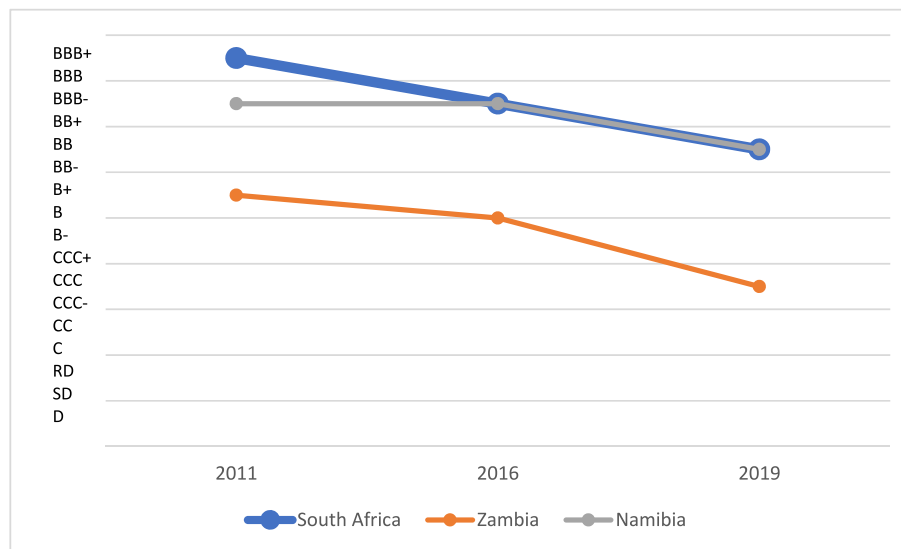


Fig. 5. Fitch sovereign credit Rating, 2011–2020: South Africa, Zambia and Namibia (source: Fitch, 2021)

technology developments (Kruger, 2022, 2023; Kruger and Eberhard, 2019). In fact, auctions were initially triggered in all three countries through either a lack of planning or a failure to translate plans into procurement, resulting in electricity supply crises. While the cases still support the importance of the planning–procurement nexus for long-term investment growth and sustainability, South Africa’s experience suggests that an indicative (as opposed to a prescriptive) planning framework is better suited to a fast-changing technological landscape where auction results routinely defy planning assumptions.

#### 4.3. Project level elements

##### 4.3.1. Equity and debt providers

While the cost of capital is a key determinant of project prices for capital-intensive renewable energy installations, the IPP literature stresses the importance of the sources of capital for project sustainability. Equity providers (sponsors) as well as debt providers (lenders) should ideally have a strong track-record – in Africa and beyond – as well as a development mandate to help them withstand and navigate short-term pressures. It has also been posited that projects developed by development-oriented sponsors are also more likely to achieve balanced outcomes (between investment return and prices), strengthening projects’ overall sustainability. Finally, IPPs should preferably be financed by long-tenored, local currency debt to reduce prices and protect off-takers from foreign currency fluctuations (Eberhard et al., 2016; Eberhard and Gratwick, 2011).

The three cases show that auctions tend to decrease the importance of the developmental mandate of project sponsors (Kruger, 2022, 2023; Kruger and Eberhard, 2019). This is mainly due to the fact that achieving balanced project outcomes is no longer determined by this developmental mandate, but rather by the competitive pressures in the auction process.

Apart from concessional financing, competition between lenders and an increase in the types of debt available reduced the cost of capital for projects. Auctions have generally proven to be quite effective at finding or creating bankable projects – at least compared to feed-in tariffs or the absence of programmatic support measures – in turn increasing competition between lenders (including DFIs) under pressure to increase their loan books. In South Africa’s case, where international DFI funding was limited, the use of corporate finance by large international bidders (as an opening negotiation position) and the entry of institutional investors (for example, pension funds) in later bid windows reduced the cost of lending as commercial banks were forced to improve their

financing terms to maintain market share.

Finally, the cases highlight the impact of lender involvement early in the auction design process as pivotal to reducing the cost of debt and increasing the effectiveness of procured projects. In all three cases, project contracts (especially PPAs) were tested (sometimes multiple times) with lenders prior to the bidding programmes being launched to ensure that they were “bankable” – that is, that they met the banks’ due diligence standards and allocated risks in a fair, transparent and consistent manner. These “tests” involved formal assessments by lender representatives and their legal advisors, as well as more informal discussions on specific provisions, to establish whether the risk allocation in these contracts are acceptable to the banks’ credit committees. This added due diligence has improved the quality of projects that are bid, and their realisation rates.

##### 4.3.2. Ensuring a secure revenue stream: off-takers, contracts and risk mitigation

Because most IPPs are project-financed and highly leveraged, it is essential that projects’ revenue streams are adequate and secure. Having a solvent, creditworthy off-taker of power is thus a key requirement for ensuring project bankability and long-term sustainability. Investors (including lenders) also require a fair allocation of risks in the project contracts (most importantly the power purchase agreement – PPA) based on the principle that the entity best positioned and equipped to manage a risk should be the one allocated that risk. Remaining risks to the revenue stream – either because of credit risks posed by, for example, an insolvent state-owned entity or because there are gaps in the contractual arrangements – need to be covered by risk mitigation and credit enhancement measures to ensure bankability. These measures include sovereign guarantees, liquidity support arrangements and multilateral, political-risk insurance products (Eberhard et al., 2016; Eberhard et al., 2017; Eberhard and Gratwick, 2011).

Unsurprisingly, the level of risk cover required for project bankability in the three cases was inversely proportional to the financial health of the off-taker and the host country (Fig. 5 and Table 5) (see Table 6).

#### 4.4. Programme-level elements: auction design

##### 4.4.1. Ensuring effectiveness

Auction effectiveness can be strengthened through using appropriate qualification criteria, penalties, and access requirements. Qualification criteria can be classified as either technical (for example, permits, land,

transmission access, equipment standards) or financial (for example, bid bonds). The aim of using qualification criteria is to ensure that only competent and committed bidders participate with well-prepared projects. According to the literature, high qualification thresholds can thus lead to higher project-realisation rates, but can also reduce competition, which can lead to higher prices (Del Río, 2017b; Diniz et al., 2023; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Kreiss et al., 2016; Matthaus, 2020; Welisch, 2018).

Auction effectiveness can also be increased through using “auction access criteria” as defined by Haufe and Ehrhart (2018) and Kreiss et al. (2016), such as requiring that bidders have a minimum level of prior experience and/or capacity, usually covering technical and financial areas. Access requirements – unlike qualification requirements – do not incur additional bid preparation costs, but the objective is the same. Staged bidding can be used to potentially decrease transaction costs for both auctioneers and bidders by screening out unqualified bidders in a prequalification stage before they submit full bids, although this can reduce competition.

The empirical evidence (Kruger, 2022, 2023; Kruger and Eberhard, 2019) suggests that setting stringent technical qualification criteria can support high project-realisation rates, but at potentially higher prices as fewer projects will be able to bid (at least in the initial auctions), thereby reducing competition. South Africa’s first auction rounds were, for example, undersubscribed, and delivered projects at high prices in large part due to the demanding qualification criteria meant to ensure the timely realisation of projects (heavily influenced by lender requirements), which meant that few projects were ready to be submitted, leading to low levels of competition. This impact was mitigated over time as more projects were prepared for later rounds, thereby increasing competition. The impact could also have been mitigated in initial rounds through the provision of prepared project sites, along with grid connections, permits and relevant data (for example, resources, geotechnical studies). This approach worked well leading to low prices and relatively short realisation timelines, though some teething issues in Zambia (Scaling Solar 1) and Namibia (Hardap) caused some delays.

The most influential qualification criteria, from an effectiveness perspective, seems to have been the requirement, in all three countries, for signed letters of support, or term sheets, from project equity investors and lenders (Table 7). Investors were unlikely to provide these support letters or term sheets without having conducted due diligence on bids, which ensured that projects were well prepared and sufficiently de-risked by the time of submission to ensure timely realisation.

Other financial qualification criteria, such as bid bonds, appear to have been less influential on auction effectiveness and efficiency. The bidding programmes used a wide range of bid bonds (Table 8), yet realisation rates remained high across the cases – and was quickest for the project with the lowest bid bond (Hardap). The impact of the levels of bid bonds on overall competition levels also appears to be negligible relative to other factors, such as the size of the auction, technical qualification criteria and the perceived capacity of the auctioneer (Kruger, 2022, 2023; Kruger and Eberhard, 2019). Nevertheless, the presence of a bid bond – irrespective of the level – appears to have an impact on the qualitative aspect of competition, with stronger bidders for example choosing to not participate in Namibia’s Khan tender due to the lack of bid bond requirements (see Appendix E).

**Table 5**

Population; GDP/capita; ease of doing business, and installed capacity - South Africa, Zambia, and Namibia (Source: IMF, 2020; The World Bank Group, 2020).

|              | Population<br>(million) | GDP (Current US<br>\$, Billion) | GDP/capita<br>(US\$ 2019) | Ave. inflation rate<br>(2010–2019) | Debt to GDP<br>ratio (2010) | Debt to GDP<br>ratio (2019) | Ease of doing<br>business score | Installed power<br>generation capacity<br>(GW) |
|--------------|-------------------------|---------------------------------|---------------------------|------------------------------------|-----------------------------|-----------------------------|---------------------------------|------------------------------------------------|
| South Africa | 58.56                   | 351.4                           | 6001.40                   | 5.17                               | 34.7                        | 62.2                        | 67.0                            | 45                                             |
| Zambia       | 17.86                   | 23.1                            | 1291.34                   | 9 <sup>a</sup>                     | 18.9                        | 91.9                        | 66.9                            | 2.8                                            |
| Namibia      | 2.49                    | 12.7                            | 4957.46                   | 5.17                               | 16                          | 65.7                        | 61.4                            | 0.6                                            |

<sup>a</sup> In 2016 inflation was 17.9%.

**Table 6**

Risk mitigation and credit enhancement measures - South Africa, Zambia, and Namibia.

|              | Offtaker<br>financial<br>health   | Sovereign<br>guarantee             | Liquidity<br>support                                                                                 | Multilateral<br>payment/loan<br>cover                                                                     |
|--------------|-----------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| South Africa | Insolvent<br>offtaker<br>(Eskom)  | Implementation<br>Agreement        | None                                                                                                 | None <sup>a</sup>                                                                                         |
| Zambia       | Insolvent<br>offtaker<br>(ZESCO)  | Government<br>Support<br>Agreement | Scaling<br>Solar: Letter<br>of credit<br><br>GETFIT:<br>Regional<br>Liquidity<br>Support<br>Facility | Scaling Solar:<br>Partial Risk<br>Guarantees<br>(payments and<br>loans <sup>b</sup> )<br><br>GETFIT: None |
| Namibia      | Solvent<br>offtaker<br>(NamPower) | None                               | None                                                                                                 | None <sup>c</sup>                                                                                         |

<sup>a</sup> Some international equity investors have taken out MIGA cover on South African REI4P projects.

<sup>b</sup> The loan guarantees were not taken up since all debt was provided by DFIs or export credit agencies.

<sup>c</sup> While a loan cover package was not officially provided as part of the bidding programme, a loan guarantee from Proparco allowed the project lender – Standard Bank – to improve the terms of its loan package.

The auction literature generally views the above qualification criteria as potentially lowering competition (i.e. number of bids) through increasing the barriers to entry (Del Río, 2017b; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Kreiss et al., 2016). While the cases illustrate that this is generally true when looking at the number of bids submitted, it is also clear that the impact on competition levels is not straightforward. All three cases illustrate that having stronger bidders as part of the bidding pool not only increases competitive pressures between bidders but also lowers prices through these bidders’ negotiating power relative to suppliers and service providers. Prices are also lowered through the ability of stronger bidders to bring more elements of the project development, construction and operations value chain in-house – thereby diversifying and increasing their sources of revenue. Such bidders may also have the ability to bid on a corporate finance basis, which lowers the cost of capital, and they have the ability to bid a portfolio of projects (if the auction is large enough), thereby reducing costs through economies of scale (Kruger et al., 2021).

The use of penalties (for example, performance bonds or liquidated damages clauses) and incentives (for example, early connection payments) are further meant to incentivise timely project realisation and improved project performance. However they can also lead to higher costs (through increasing risks) and lower competition (by acting as barriers to entry), thereby increasing prices (Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Kreiss et al., 2016; Kruger et al., 2021; Matthaus, 2020). All three countries’ programmes included strict, and relatively severe, penalty regimes in the auctions’ designs (Table 9), but the effectiveness of these arrangements is not clear. In general, there

**Table 7**

Access requirements and staged bidding: South Africa, Zambia, and Namibia.

|              | Reference projects                                                                                                                                                                                         | EPC & O&M                                                                    | Min. turnover/<br>net assets                                                                                                                                                    | Staged bidding                       |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| South Africa | None                                                                                                                                                                                                       | Heads of agreements and reference projects                                   | Net assets of all finance providers must be 100% of what they propose in past 3 years<br>OR Proven ability to raise financing similar to what is being proposed in past 5 years | REI4P: one-stage<br>SPI4P: two-stage |
| Zambia       | Scaling Solar: 25 MW PV or 75 MW any tech in Africa; or 3x PV (100 MW each) or 1 × 1500 MW any tech<br>GETFIT: 2 × 5MW PV in Africa; or 100 MW PV in Africa; or 500 MW PV globally; or 750 MW RE globally. | Scaling Solar: Heads of contracts and reference projects<br>GETFIT: As above | Scaling Solar: Net worth US \$75 million<br>GETFIT: Total Assets US\$25 million                                                                                                 | Scaling Solar &<br>GETFIT: Two-stage |
| Namibia      | 1 × 10 MW solar PV (Hardap)<br><br>1 × 5 MW solar PV (Khan)                                                                                                                                                | Hardap: Heads of agreements and reference projects<br>Khan: None             | None<br>None                                                                                                                                                                    | Hardap: One-stage<br>Khan: as above  |

appears to have been a reluctance to impose penalties when problems were experienced, with auctioneers not wanting to jeopardise the realisation of distressed projects by unnecessarily increasing pressure (Kruger, 2022, 2023; Kruger and Eberhard, 2019). Nevertheless, the existence of the penalties may have impacted initial bidder behaviour by for example improving project quality or project management – despite not being used in the end – thereby improving effectiveness.

#### 4.4.2. Setting and dividing the auction volume

Setting auction volumes, project size limits, technologies and spreading them across time is an important part of the auction design process that impacts competition, costs and effectiveness. Volumes should ideally be based on a least-cost, regularly updated, rational power expansion plan that matches demand with supply on a least-cost basis (Eberhard et al., 2017). Theoretically, large auction volumes can attract more competition (Ballesteros-Pérez et al., 2016; Friedman, 1956), but if volumes are set too high, the auction runs the risk of being undersubscribed, resulting in low competition levels and high prices.

**Table 8**

Financial and physical qualification requirements - South Africa, Zambia, and Namibia.

|              | Bid Bonds (US \$/MW)                      | Site/land agreements required at submission                                                                        | Permits required at submission                                                                      | Transmission connection at bid submission                                                     | Lender/investor letter of support at submission                                                            |
|--------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| South Africa | 8000                                      | Lease agreement/title deed                                                                                         | EIA; aviation; water use licence; heritage; zoning application                                      | Cost estimate letter from Eskom                                                               | Binding letters of support (debt and equity)                                                               |
| Zambia       | 30,000 (Scaling Solar)<br>15,000 (GETFIT) | Scaling Solar: site provided<br>GETFIT: conditional lease agreement/title deed/undertaking (10 km from substation) | Scaling Solar: EIA provided. No others required at submission.<br>GETFIT: draft environmental brief | Scaling Solar: Transmission provided<br>GETFIT: no formal grid connection agreement required. | Scaling Solar: Signed term sheets<br>GETFIT: Signed term sheets (non-binding)                              |
| Namibia      | 800 (Hardap)<br><br>0 (Khan)              | Hardap: site provided<br>Khan: site selected, but not secured                                                      | Hardap: EIA provided<br>Khan: none required at submission                                           | Hardap: provided<br>Khan: provided, but at cost to project                                    | Signed term sheets (non-binding)/letters of support<br>Signed term sheets (non-binding)/letters of support |

Auctioneers can also decide to limit the amount of capacity or energy that can be awarded to a single bidder or project. Such limits can increase the costs of energy through limiting economies of scale but are often included owing to energy security considerations – meaning that there is a desire to spread the risk of project non-realisation across a number of entities – and desire to diversify power generation. Auction volumes should also, ideally, be bid out over a number of subsequent rounds (Eberhard et al., 2014; Ferroukhi et al., 2015); this has been shown to increase competition and lessen the impact of unsuccessful bids by providing these bidders with the opportunity to resubmit bids in later rounds.

South Africa's experience in the first rounds of bidding is a classic example of large auction volumes resulting in reduced competition levels and higher prices (Kruger, 2023). Auction volumes in Zambia and Namibia were prudently set at much lower levels, resulting in higher levels of competition.

Beyond smaller auction volumes, the use of multiple auction rounds was crucial to driving down prices – or, at least, capturing cost decreases – in all three case-study countries (Kruger, 2022, 2023; Kruger and Eberhard, 2019). This re-emphasises the importance of the planning-procurement nexus – not only for securing investment but crucially also for capturing and driving cost reductions.

#### 4.4.3. Determining winners

Key elements in the design of auctions include the auction format (sealed bid vs. descending clock), the pricing rule (pay-as-bid vs uniform pricing), and the criteria used for bid evaluation and scoring. Auctioneers also need to decide whether to set, and whether to disclose, ceiling prices, also called price caps or reserve prices. Ceiling prices, generally protect the auctioneer (and, therefore, also electricity consumers) from securing projects at prices that are considered too high. Setting ceiling prices too low can reduce the number of bids being submitted (but strengthen the quality of the remaining bidding pool) or lead to no, or very few, projects being awarded, which can, in turn, threaten energy security.

A sealed-bid auction format offers fewer opportunities for collusion since bidders cannot signal to each other through the bidding process, but a descending clock auction can, theoretically, increase the realisation rate of projects since bidders are better able to benchmark their costing assumptions against their competitors (Haufe and Ehrhart, 2018; Hubbard and Paarsch, 2016; Klemperer, 2004a).

All three case-study countries elected to make use of the simpler, more straightforward sealed-bid, pay-as-bid approach. Only South Africa used price caps, and then only in the first three rounds of the REI4P. Bidders tended to bid close to these levels, rather than push lower price boundaries.

To determine winning projects, bids can be scored on price only (generally regarded as a theoretically more efficient option) (Estache and Iimi, 2012), or on the basis of other criteria alongside price, such as socio-economic development commitments or technical quality (for example, project preparedness, quality of equipment, and so on) to

**Table 9**

Penalties and incentives - South Africa, Zambia, and Namibia.

|              | Performance bonds (US \$/MW)                   | Liquidated damages                                                                                                                                                                         | PPA length reduction                           | Early connection payment     |
|--------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------|
| South Africa | 16,000                                         | None, though projects were awarded termination points for SED underperformance                                                                                                             | 3 days' reduction for 1-day delay              | 60% of Bid tariff            |
| Zambia       | Scaling Solar: 300,000                         | Scaling Solar: US\$7500 for every 0,1% annual plant performance ratio below estimated performance ratio                                                                                    | Scaling Solar: 1 day reduction for 1-day delay | Scaling Solar: 80% of tariff |
|              | GETFiT: None                                   | GETFiT: US\$500/MW per-day delay                                                                                                                                                           | GETFiT: As above                               | GETFiT: As above             |
| Namibia      | Hardap: 2% EPC (up to FC); 15% EPC (up to COD) | Hardap: 0,25% of EPC costs/week delay<br>1% of EPC costs for every 0,1% below contracted performance ratio<br>1% of EPC costs for every 100 kW of which plant is below contracted capacity | Hardap: 1 day reduction for 1-day delay        | None                         |
|              | Khan: None                                     | Khan: None                                                                                                                                                                                 | 1 day reduction for 1-day delay                | None                         |

incentivise additional performance on these metrics beyond those stipulated as the minimum qualification criteria (Ferroukhi et al., 2015; Haufe and Ehrhart, 2018) (Table 9). In general, the use of additional awarding criteria is fraught with potential problems, such as a lack of transparency or sub-optimal outcomes (Estache et al., 2009). Nevertheless, only the Scaling Solar programme amongst the case studies chose to make use of price as the sole awarding criterion (Table 10).

The consequences of employing additional criteria in deciding bid winners were mixed (Kruger, 2022, 2023; Kruger and Eberhard, 2019). The use of technical criteria, such as project performance or realisation timelines, as additional awarding criteria appears to have delivered little additional impact. The Hardap project was the single project to include technical bid scoring criteria such as capacity factors and project schedules. Results indicate that these criteria did not have an additional impact on project performance since the incentives already in place through the general contractual framework were strong. GETFiT used a combination of technical and socio-economic development criteria, but at least two of these ("scheduled COD as per PPA" and "project readiness") have been rendered largely inconsequential due to the delays in solving the programme's general bankability issues. GETFiT's use of projects' contribution to system losses is, however, a potentially important technical evaluation criterion that is likely to be used more frequently as countries grapple with integrating larger shares of renewables in their power systems.

The use of socio-economic development (SED) evaluation criteria appears to have delivered more tangible results, and is becoming more widespread in the region. Projects' commitments on key SED metrics in SA's REIP have generally increased over the bidding rounds to levels at, or beyond, the targets<sup>7</sup> set by the IPP office, leading to more socio-

<sup>7</sup> The REIP RFP specified both threshold levels (that is, qualification criteria) and targets for SED indicators. Targets differed from thresholds in that not meeting them would not disqualify a project, while meeting them would result in a project being awarded the maximum number of points (at least in the initial rounds).

**Table 10**

Bid evaluation and scoring criteria: South Africa, Zambia, and Namibia.

|              | Price                            | Other                                                                                                                                                                                                                                                                                                                        |
|--------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| South Africa | 70                               | 30<br>Job creation (25%)<br>Local content (25%)<br>Ownership (15%)<br>Management control (5%)<br>Preferential procurement (10%)<br>Enterprise development (5%)<br>Socio-economic development (15%)                                                                                                                           |
| Zambia       | Scaling Solar: 100<br>GETFiT: 80 | GETFiT: 20<br>Project readiness (40%)<br>Scheduled COD as per PPA (20%)<br>Contribution to total system losses (25%)<br>Local capacity-building and training (15%)                                                                                                                                                           |
| Namibia      | Hardap: 70<br><br>Khan: 80       | Hardap: 30 – Technical<br>Total degradation factor Year 25 (33%)<br>Capacity factor Year 2 (33%)<br>Project schedule to COD (33%)<br>Khan: 20 – Socio-economic development<br>% Revenue to community development (25%)<br>PDN in executive management (25%)<br>Procurement spending as % of EPC value on PDN companies (50%) |

economic development benefits for the country and local communities as opposed to what would have been the case had these criteria been used only as qualification criteria. However, there are questions as to how well these commitments have translated into actual results (Baker and Sovacool, 2017; Davies and Morar, 2016; Marais et al., 2018; Stands, 2015; Wlokas, 2015; Wlokas et al., 2012), whether they have allowed for gaming of the system and what the impact has been on pricing – and as such requires further investigation.

#### 4.5. Programme level elements: implementation

Beyond the auction design elements, the literature and the empirical cases further foreground the importance of trust – in the auctioneer and in the bidding process – as a determinant of competition levels and the effectiveness of auctions. The auctioneer also functions as coordinator, or at least as a neutral third party (honest broker) to help projects reach financial close and commercial operation deadlines.

##### 4.5.1. Trust in the bidding authority

To earn bidder trust, the institution designing and implementing the auction should ideally be well resourced and politically supported, capable, and seen as having integrity (Eberhard et al., 2014; Meyer et al., 2018). The implementing team should ideally also be an agile and capable professional group, committed to institutional innovation (Boulle et al., 2015; Garud et al., 2007; Najam, 1995).

The importance of trust in the bidding authority is evident in all three cases. The identity of the auctioneer, as much as any of the other factors already analysed, influenced competition levels (especially the quality of bidders) and resulting prices in all three countries (Kruger, 2022, 2023; Kruger and Eberhard, 2019). In some cases, such as GETFiT Zambia, it seemed to be the determining factor in price-setting (Kruger and Eberhard, 2019).

Trust in the bidding authority is determined by the auctioneer's perceived capability, integrity and benevolence (Mayer et al., 1995). In all three cases, auctioneers (or supporting institutions) had a track record of successful collaboration with the private sector, and were generally seen as trustworthy.

The cases also provide evidence of the auctioneers playing important coordination and brokerage roles to get projects over the line (Kruger, 2022, 2023; Kruger and Eberhard, 2019). South Africa's IPP office played a centralised coordination role in the post-award period, helping projects navigate the complexity of the various permitting and other



regulatory approval processes. In Zambia, the IDC played a muted role in getting awarded projects to financial close, mainly due to internal ructions in the organisation, but still helped resolve issues around for example, land ownership. Bidders however lamented the lack of a capable, trusted intermediary, such as the IFC, in the post-award period to get projects across the line. Still, the IDC, with the president as chairman, was a powerful ally of the programme that did a great deal of path-clearing work. In the case of GETFIT, the Ministry of Energy appears to have been a less capable and supportive partner, seemingly incapable or unwilling to remove the roadblocks and, at times, completely locked out of the energy sector decision-making processes. In Namibia, NamPower played an important role in getting projects closed and built, not only as procurer but also as shareholder, resulting in the shortest project financial close and construction period among the programmes.

#### 4.5.2. Trust in the bidding process

Bidders do not only need to trust the bidding authority, but also the bidding process. This trust is based on perceptions of the integrity, security, transparency and justice of the bidding process – in short, whether the programme and its outcomes are seen as transparent, fair and secure (Chiu et al., 2010; Zitron, 2006).

All three countries' programmes appear to have sent the right signals to secure the market's trust as the first rounds of bidding were launched (Kruger, 2022, 2023; Kruger and Eberhard, 2019): RFP documentation was generally regarded as well-developed and robust; auction rules were seen to be clear and mostly fair; bid clarification questions were transparently answered and widely circulated, and bid opening and evaluation processes were secured through CCTV security, independent observers and/or investment committees. South Africa went the furthest in its efforts at setting up a fair and transparent auction process, in part to protect the integrity of the auction and its outcomes (especially from politically connected rent-seekers), but also to secure and maintain the trust of a sceptical market that had been gearing up for an aborted REFIT process.

#### 4.6. The modified analytical framework

The result of the preceding comparative case study analyses (incl. Kruger, 2022, 2023; Kruger and Eberhard, 2019) is the development of a modified analytical framework (Table 3) that incorporates the empirical findings and allows for a more accurate and thoroughgoing analysis of the factors impacting renewable energy auctions price and project realisation outcomes in Africa and other developing regions. Text in blue denotes modifications to the original framework (Table 11) based on the empirical analysis.

Modifications at country level largely illustrate the impact of the broader enabling environment on competition levels, while specifically also highlighting the impact of competitive procurement practices on project realisation. At an auction design level, many of the modifications clarify the non-impact of certain design choices on project realisation, highlight the impact of investor commitment as qualification criterion on project realisation, clarify the differences in impact on the quantity vs. quality of competition of certain design choices and show that price caps can increase (rather than decrease) project prices. In terms of auction implementation, the modifications focus on trust in the auctioning entity and auction process as improving project realisation, while adding auctioneer assistance post-award as influential in increasing competition levels. At a project level, the modifications show that auctions generally dilute the influence of investors' developmental mandate in comparison with their overall size and financial standing, illustrate how the incorporation of lenders' needs and commitments strengthen project realisation and highlight the potential positive impact of local strategic shareholding on project realisation.

## 5. Conclusion and policy implications

The analysis has shown that renewable energy auction project price outcomes appear to be mainly influenced by two factors: competition levels and cost of capital.<sup>8</sup> This is not necessarily a new finding, merely confirming what others in the field have noted before (Dobrotkova et al., 2018; Gephart et al., 2017b; Haufe and Ehrhart, 2018; Steffen, 2020). However, this paper provides deeper insight into the factors influencing both competition levels and cost of capital, particularly in the African context.

Competition levels are determined by both the quality or strength of bidders – that is, their experience, size and access to finance – and the quantity of bidders, and are influenced mainly by:

- the volume being procured, with prudent, smaller volumes<sup>9</sup> increasing competition levels;
- whether the auction is designed and perceived as a credible, relatively low-risk programme, implemented by a trusted, credible authority that limits participation to credible bidders; and
- whether bidders are assured of a secure project revenue stream through the provision of bankable contracts with creditworthy off-takers, including fair risk allocation and mitigation provisions such as credit enhancement and security measures (where necessary).

Notably, beyond the existence of a procurement programme as a credible route to market and a policy framework that allows for private power investments, many of the country-level elements identified by the IPP literature do not appear to necessarily influence competition levels.

Cost of capital – the pricing of risk – is normally determined by summing country, programme and project risks. Country risk is impacted by factors such as the investment environment, the quality of regulation and the creditworthiness of the offtaker utility. Project risks are also key – and lenders will do a detailed due diligence of individual projects: of the strength and quality of the bidder; of the adequacy and security of the revenue stream which repays debt and rewards equity, and of the available credit enhancement and security measures. The analysis has shown, however, that the programme level – that is, how an auction is designed and implemented – plays a further role through increasing competition levels, inducing bidders to reduce margins along the value chain (including cost of capital); and that trust in the auction implementing authority can supersede the country risk factors in determining the cost of capital.

Auction designers could therefore increase the likelihood of efficient price outcomes by ensuring that well-designed procurement programmes are implemented by respected, capable, supported institutions, and that key project contracts are considered bankable.

The size and costs of debt finance – as a subset of the cost of capital – have also been shown to be particularly important for determining project prices. High gearing ratios, concessionary interest rates and long

<sup>8</sup> Prices (measured as the levelized cost of electricity) from renewable energy projects are mainly impacted by three factors: investments costs (cost of equipment, construction, operations, including land, labour, permits), cost of capital (equity and debt), and capacity factors (the amount of energy that can be generated by the plant) (Bazilian et al., 2013; Dobrotkova et al., 2018). Prices have obviously been impacted by the drop in prices of solar PV panels and other equipment, but we argue that without sufficient levels of competition, these cost benefits are less likely to be discovered and captured (as evidenced by the higher prices for directly negotiated and feed-in tariff projects). Capacity factors have been shown to be less influential than cost of capital, and solar resource levels between the three countries are relatively similar (Dobrotkova et al., 2018; Frisari and Stadelmann, 2015; Labordena et al., 2017; Ondraczek et al., 2015; Steffen, 2020).

<sup>9</sup> Volumes still need to be big enough to attract bids – especially from stronger bidders (Ballesteros-Pérez et al., 2016; Friedman, 1956). Project sizes need to be at least 20 MW or above to avoid significant scale-cost trade-offs.

**Table 11**

Modified analytical framework - with price and realisation impacts (+ denotes a positive impact on the outcome; - denotes a negative impact; = denotes no clear impact).

| Factors                                                   | Detail                                                                                                                                                                                                                                                                            | Impact on Price                                                                                                                                                                    | Impact on Realisation                                                                                                                    |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Country level                                             |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                    |                                                                                                                                          |
| Stability of economic and legal context                   | Stability of macroeconomic policies (incl. repayment record & investment rating)<br>The legal system allows contracts to be enforced, laws to be upheld, and arbitration to be fair                                                                                               | (+) Stable investment context lowers investment risks, reducing cost of capital and increasing competition levels, leading to lower prices <sup>a</sup>                            | (+) Stability of investment context – especially judicial quality – improves contractual effectiveness and thereby realisation           |
| Energy policy framework                                   | Energy policy framework allows private power investment<br>Market access through structured (preferably competitive) procurement programmes                                                                                                                                       | (+) Clear energy policy that allows for private power investment reduces investment risks and increases competition levels, leading to lower prices                                | (+) Clear energy policy that allows for private investment improves project realisation                                                  |
| Regulatory transparency, consistency and fairness         | Transparent and predictable licensing and tariff framework (preferably included in bidding requirements/guidance)<br>Cost-reflective retail tariffs                                                                                                                               | (+) Good regulatory quality lowers investment risks, reducing cost of capital and increasing competition, leading to lower prices                                                  | (+) Good regulatory quality, especially on licensing and tariff approval, improve project realisation                                    |
| Coherent sectoral planning                                | Power-planning roles and functions clear and allocated to skilled, resourced, empowered planners<br>Fair allocation of new-build opportunities between utilities and IPPs<br>Plans are indicative rather than determinative                                                       | (+) Coherent sectoral planning reduces investment risks, reducing cost of capital, and increases competition, leading to lower prices                                              | (+) Coherent sectoral planning improves project pipeline development, improving project realisation                                      |
| Competitive bidding practices                             | IPP procurement happens through competitive bidding programmes<br>Planning linked to timely initiation of adequately resourced competitive tenders/auctions – although auctions can be initiated outside planning framework if necessary                                          | (+) Competitive bidding practices lower project prices                                                                                                                             | (+) Efficient competitive bidding practices shorten project development and negotiation timelines, improving realisation                 |
| <b>Programme level</b>                                    |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                    |                                                                                                                                          |
| Auction Design                                            |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                    |                                                                                                                                          |
| Competition-inducing, scale-friendly auction volume       | Auction volumes large enough to attract bidders (20 MW+), but small enough to ensure competitive pressure (especially in early rounds) - and preferably based on least-cost planning framework                                                                                    | (++) Auction volume that is large enough to attract bidders, but smaller than offered volume, induces competition, leading to lower prices                                         | (=) No theoretical expectation around impact on realisation                                                                              |
| Scale-friendly project size limits                        | Project size limits – if used – are based on clear market development, energy security criteria & enable economies of scale                                                                                                                                                       | (–) Project size caps limit economies of scale, increasing prices <sup>b</sup>                                                                                                     | (+) Smaller projects are more quickly built, improving realisation                                                                       |
| Technology bands                                          | Initial auction rounds based on simple, tested technologies<br>Auction allows for adjustment of technology limits based on price and system optimisation considerations                                                                                                           | (–) Technology bands limit competition, increasing prices                                                                                                                          | (=) No theoretical expectation around impact on realisation                                                                              |
| Bid award limits                                          | Bid award limits – if used – set to counter legitimate market domination or energy security concerns                                                                                                                                                                              | (–) Bid award caps limit economies of scale, increasing prices                                                                                                                     | (=) Bid award limits appear to have no discernible impact on realisation                                                                 |
| Bidding rounds                                            | Regular, scheduled bidding rounds increase competition and capture learning rates                                                                                                                                                                                                 | (++) Bidding rounds increase competition and induce market learning, lowering prices                                                                                               | (+) Bidding rounds increase project pipeline development and market learning, improving realisation <sup>c</sup>                         |
| Multi-criteria winner selection                           | Winner selection criteria transparent and rational<br>Price key (preferably only) determinant<br>Additional scoring criteria focus on socio-economic or system level impacts                                                                                                      | (–) Multiple bid scoring criteria dilute the importance of price and add additional cost drivers, increasing prices <sup>d</sup>                                                   | (=) Technical & SED scoring criteria appear to have no discernible impact on realisation                                                 |
| Price caps (unknown)                                      | Price caps (ceiling prices) undisclosed, if used                                                                                                                                                                                                                                  | (–) Undisclosed or no price caps discourage price anchoring, lowering prices                                                                                                       | (=) No theoretical impact expectation, although price caps being set too low can result in no project being awarded                      |
| Physical (project/bid preparation) qualification criteria | Physical (project/bid preparation) qualification criteria include, as a minimum, investor (lender) commitment<br>Late-stage qualification criteria coupled with long bidding timelines (for initial rounds) or site provision and preparation (incl. grid, permits) by auctioneer | (±) High physical qualification criteria increase bidding costs, lowering number of bids but increasing quality of bids (and if provided by auctioneer, can lower costs and risks) | (++) High physical qualification criteria (esp. lender commitment) improve project preparation and cost certainty, improving realisation |
| Financial (bid bond) qualification criteria               | Bid bonds set to ensure commitment of qualified bidders and the quality of competition                                                                                                                                                                                            | (±) Bid bonds increase bidding costs, lowering the number of bids but increasing the quality of bids                                                                               | (+) Bid bonds increase bidder commitment and cost certainty, improving realisation                                                       |
| Auction access requirements                               | Auction access requirements – technical experience or financial status – set to limit participation to experienced, capable bidders                                                                                                                                               | (±) High auction access requirements lower number of bids, but improve quality of bids.                                                                                            | (+) High auction access requirements increase likelihood of capable bidders being awarded, increasing realisation                        |
| Penalty regime                                            | Reasonable penalties (PPA reduction, performance bonds) incentivise project completion<br>Liquidated damages used to incentivise performance on non-technical aspects (e.g. SED commitments)                                                                                      | (–) Penalties increase project risks, increasing the cost of capital and other project costs, increasing prices                                                                    | (+) Penalty regime incentivises project preparation and cost certainty, improving realisation                                            |
| Staged bidding process                                    | Choice of one- vs. two stage bidding process balances transaction costs with time concerns                                                                                                                                                                                        | (–) Staged bidding reduces competition, increasing prices                                                                                                                          | (=) No clear evidence of impact of staged bidding on realisation                                                                         |
| Pay-as-bid, sealed bid format                             | Simple sealed bid, pay-as-bid auction format used to reduce bidder risks and increase competition levels                                                                                                                                                                          | (+) Simple bidding format and pricing rule increases competition, leading to lower prices                                                                                          | (+) Simple bidding format reduces winner's curse risk, improving realisation                                                             |
| <b>Auction Implementation</b>                             |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                    |                                                                                                                                          |

(continued on next page)

Table 11 (continued)

| Factors                                        | Detail                                                                                                                                                                                                                                                                                                                                         | Impact on Price                                                                                                                      | Impact on Realisation                                                                                                                                     |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Procurer                                       | Procuring entity perceived as capable, benevolent and acting with integrity to build and maintain bidder trust                                                                                                                                                                                                                                 | (++) Trust in the procuring authority increases competition, lowering prices                                                         | (+) Trust in procuring entity attracts capable, experienced bidders, improving realisation                                                                |
| Procurement process                            | Bidding process is perceived as fair, transparent, secure to build and maintain bidder trust                                                                                                                                                                                                                                                   | (++) Trust in bidding process increases competition, lowering prices                                                                 | (+) Trust in procurement process attracts capable, experience bidders, improving realisation                                                              |
| Support to awarded projects                    | Auctioneer assists awarded projects as honest broker to reach financial close and commercial operation                                                                                                                                                                                                                                         | (+) Auctioneer assistance improves likelihood of realisation, improving competition and lowering prices                              | (+) Auctioneer support to awarded projects to financial close and COD improves realisation                                                                |
| <b>Project level</b>                           |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                      |                                                                                                                                                           |
| Favourable equity partners                     | Experienced (possibly international) firms with strong balance sheets compete to secure project contracts and realise projects<br>Local strategic shareholding encouraged                                                                                                                                                                      | (+) Strong, experienced equity partners can lower project costs and cost of capital, lowering prices                                 | (+) Experienced, strong equity partners able to navigate realisation risks, improving realisation<br>(+) Local strategic shareholders improve realisation |
| Favourable debt arrangements                   | Competitive (low cost, long-tenored) debt – preferably with considerable local capital to mitigate foreign exchange risk<br>Lenders review project contracts/bidding documents to ensure bankability                                                                                                                                           | (+) Favourable debt arrangements lowers the cost of capital, lowering prices                                                         | (+) Debt from DFIs come with a halo effect, improving realisation<br>(+) Auction design in line with lender requirements improve realisation              |
| Secure revenue stream                          | Secure revenue stream ensured through robust, bankable contracts (incl. international arbitration if necessary) with investment-grade utility (with full cost-recovery and high billing collection rates) or the use of risk mitigation and credit enhancement measures (e.g. sovereign guarantee, liquidity support, multilateral risk cover) | (++) Secure revenue stream reduces project risks, and thereby the cost of capital, and increases competition levels, lowering prices | (++) Secure revenue stream enables project to secure funding, improving realisation                                                                       |
| Strategic management and relationship building | Project bids include socio-economic development commitments – specifically strategic local shareholding (e.g. offtaker, empowerment shares)                                                                                                                                                                                                    | (–) Strategic socio-economic development commitments increase project costs & risks, increasing prices                               | (+) Strategic socio-economic development commitments improve project realisation                                                                          |

<sup>a</sup> The impact of the investment context on pricing can be moderated through a well-designed, structured auction programme that has a secure revenue stream and is implemented by a trusted auctioning entity.

<sup>b</sup> The size of this impact seems to be somewhat limited by the fact that solar PV is a modular technology that appears to scale down relatively easily and cost-effectively (at least down to 20 MW).

<sup>c</sup> The overall impact of bidding rounds on realisation is nonetheless impacted by other external (economic, political) factors as well. It is often the path-breaking rounds or projects that appear to have an easier time getting built – due to multiple reasons discussed in chapter 3–5. And 6.

<sup>d</sup> The impact of this depends on the type and quantum of criteria. In general, SED criteria seem to have a more pronounced impact than technical criteria.

loan tenors (especially for commercial loans) have been key to driving down the cost of capital and, consequently, project prices across the auction programmes. For this reason (among others), the deliberate incorporation of lender requirements as well as the testing of project documents with lenders have been highlighted as particularly important elements of auction design in the sub-Saharan African context. Competition between lenders, the availability of more sources of debt and complementary debt provision and risk mitigation arrangements between commercial lenders and DFIs also tend to lower the cost of capital through inducing lower pricing of debt.

Price is one key outcome; the other is project realisation – i.e. whether projects are actually built and reach commercial operation. The effectiveness of auctions in achieving timely project realisation is primarily impacted by project preparation levels, investor commitment (equity and debt), bidder quality, local shareholding by strategic local partners, robust risk mitigation measures, auctioneer assistance, the investment climate and the quality of regulation.

The more advanced a project is at the point of bidding, the sooner it is likely to reach financial close and commercial operation (see section 4.4.1). The analysis has shown that the most influential qualification requirement that increases project preparation levels and ensures auction effectiveness appears to be lender commitment. Lenders are potentially carrying the most risks during projects' construction and operation phases, providing the majority of project funding (upwards of 70%), often on long-term (15+ year) bases. By requiring lenders to provide signed letters or term sheets that confirm their acceptance of the project documents (including contracts) and their willingness to provide finance on these terms, auctioneers are ensuring that the most significant and most risk-averse party to the transaction has already conducted

thorough due diligence by the time of bidding.

Bidder quality – that is, the size, experience, access to resources (especially finance) and commitment of bidders – is another determinant of auction effectiveness since stronger bidders have generally been able to draw on their experience, financial (and other) resources and their relatively strong negotiating positions to ensure that projects reach commercial operation milestones, even in increasingly challenging circumstances.

A politically supported, capable auctioneer that also acts as broker and coordinator between bidders (including their investors) and the various public stakeholders (for example, ministries, regulators, utilities) in the crucial periods between bid award and financial close and commercial operation can furthermore greatly increase the effectiveness of an auction programme and is partly determined by the institutional quality underpinning the auction programme (Boulle et al., 2015; Marquard, 2006).

Penultimately, while maintaining the largely risk-based focus of the IPP literature, the analytical framework introduces an important risk-mitigating element from the auction implementation literature: trust in a respected, capable and resourced auction implementing agency and fair bidding processes. It is also through the lens of trust and its role in the bidding decision that the competition-enhancing influence of multilateral- and development-finance institutions as auctioneers (or, at least, as strategic auction implementation partners) is best understood. The focus on the influence of trust on the bidding decision – as determined by the auctioneers' perceived ability, integrity and benevolence (Chiu et al., 2010; Liu et al., 2016; Mayer et al., 1995) as well as the perceived justice of the bidding process (Chiu et al., 2010) – provides the conceptual language and a potential causal explanation for a

phenomenon that had been noted in the empirical renewable energy auction literature. It is also through the lens of trust and its role in the bidding decision (Chiu et al., 2010; Liu et al., 2016; Zitron, 2006) that the competition-enhancing influence of multilateral- and development-finance institutions as auctioneers (or, at least, as strategic auction implementation partners) is best understood. This adds further detail to the auction design best practice claims made by Del Río (2017b), Gephart et al. (2017) and Haufe and Ehrhart (2018) – none of which include auction implementation factors<sup>10</sup> in their recommendations.

Finally, the paper highlights the roles of three groups of actors for determining auction outcomes. The first group – lenders – have proven to be particularly influential in determining IPP outcomes, not merely as providers of finance, but as key stakeholders in auction design decisions. The study also provides useful nuance to the discussion around the influence of local shareholders for IPP sustainability, showing that it is not merely the nationality of the shareholders but rather their strategic influence (for example, as offtaker and/or procurer, or politically-influential empowerment investor) that helps increase auction effectiveness. The third group of actors are the auctioneers, in particular institutional champions that have the capacity, autonomy and political and financial backing to set up and run the procurement programmes in such a way that they inspire trust and ensure effective and efficient outcomes.

The study found that renewable energy auction outcomes in South Africa, Zambia and Namibia are attributable to a mixture of elements primarily drawn from a framework that combines the literature on contributing elements to IPP success – mainly focusing on the country and project levels of analysis – with the literature on renewable energy auction design and implementation (mainly focusing on trust in the auctioneer and process) – also referred to as the “programme” level. This combined analytical framework developed a suite of contributing elements to success at all three levels – country, programme and project -

that both illuminated the individual and combined contributions as well as the interplay between elements.

This analysis has been primarily based on qualitative case study analyses. As such, there are inherent limitations to the findings – especially with regards to the comparative extent of influence of each of the contributing elements discussed in sections 4.2. to 4.4. A quantitative assessment of these elements and their correlation with outcomes will assist in testing and strengthening these findings, as also pointed out by other leading scholars in the field (del Río and Kiefer, 2023).

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## CRediT authorship contribution statement

**Wikus Kruger:** Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Visualization, Project administration, Funding acquisition. **Anton Eberhard:** Supervision, Writing – review & editing, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wikus Kruger reports financial support was provided by United Kingdom Department for International Development.

## Data availability

The data that has been used is confidential.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2023.113572>.

## Appendix A. List of Interviews

| Interview # | Organisation type    | Countries covered   | Key Informant       | Review |
|-------------|----------------------|---------------------|---------------------|--------|
| 1           | Implementing partner | Zambia              | Key informant       | Review |
| 2           | RE project developer | SA, Namibia         |                     |        |
| 3           | Private equity       | SA, Namibia         | Key informant       | Review |
| 4           | Regulator            | Namibia             |                     |        |
| 5           | Utility              | Namibia             |                     |        |
| 6           | Consultant           | Namibia             |                     |        |
| 7           | Utility              | Namibia             |                     |        |
| 8           | RE project developer | SA, Namibia, Zambia |                     |        |
| 9           | RE project developer | Namibia, Zambia     |                     |        |
| 10          | Ministry             | Namibia             |                     |        |
| 11          | Regulator            | Zambia              |                     |        |
| 12          | RE project developer | Zambia              |                     |        |
| 13          | Implementing partner | Zambia              | SA, Zambia, Namibia |        |
| 14          | DFI                  | Zambia              |                     |        |
| 15          | RE project developer | SA                  |                     |        |
| 16          | RE project developer | SA, Zambia, Namibia |                     |        |
| 17          | Implementing partner | Zambia              |                     |        |
| 18          | RE project developer | SA                  |                     |        |
| 19          | RE project developer | SA, Zambia, Namibia |                     |        |
| 20          | RE project developer | SA                  |                     |        |

(continued on next page)

<sup>10</sup> Del Río (2017b) refers to “improving administrative procedures”, “supporting dialogue with stakeholders” and “information provision” as “participation conditions” that could arguably fall within the “project implementation” category, but without the explicit identification of trust being the mediator of these elements’ influence on competition levels.

(continued)

| Interview # | Organisation type    | Countries covered   | Key Informant | Review |
|-------------|----------------------|---------------------|---------------|--------|
| 21          | RE project developer | SA                  |               |        |
| 22          | EPC                  | SA, Zambia          |               |        |
| 23          | RE project developer | SA                  |               |        |
| 24          | Procurement office   | SA                  |               |        |
| 25          | Ministry             | SA                  |               |        |
| 26          | RE project developer | SA, Zambia, Namibia |               |        |
| 27          | RE project developer | SA, Namibia         |               |        |
| 28          | Implementing partner | Zambia              |               |        |
| 29          | Implementing partner | Zambia              |               |        |
| 30          | Procurement office   | SA                  | Key informant | Review |
| 31          | Implementing partner | Zambia              | Key informant | Review |
| 32          | Procurement office   | Zambia              |               |        |
| 33          | Procurement office   | Zambia              |               |        |
| 34          | Insurance agency     | Zambia              |               |        |
| 35          | Utility              | Zambia              |               |        |
| 36          | Procurement office   | SA                  |               |        |
| 37          | Regulator            | Namibia             | Key informant | Review |
| 38          | Utility              | Namibia             |               |        |
| 39          | RE project developer | SA                  |               |        |
| 40          | Private equity       | SA, Zambia          |               |        |
| 41          | Private equity       | SA, Zambia, Namibia |               |        |
| 42          | Utility              | Namibia             |               |        |
| 43          | RE project developer | SA, Zambia, Namibia |               |        |
| 44          | DFI                  | Namibia             |               |        |
| 45          | DFI                  | Zambia              |               |        |

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